

v1.0Scalar Boson Deep Dive Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Scalar Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: v1.0Scalar Boson Deep Dive Request V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{17} + particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Dark energy scalar	$\lambda = \text{const}$ assumed	DESI 2024: evolving dark energy $w(z)$ mildly preferred. QUINTESSENCE_SCALAR added with $w = w_0 + w_a(1-a)$	HIGH	Observational hint for dynamical dark energy — changes cosmological background
Swampland conjecture	Not covered	Ooguri et al. 2024: de Sitter Swampland — challenges KKLT stabilization. V2.0: acknowledgement of theoretical uncertainty	MEDIUM	Core string theory framework for scalar potential under scrutiny
Higgs inflation model	General inflation	Planck 2023: $r < 0.036$, $n_s = 0.965$ — Higgs inflation predicts $r \sim 0.003$ within future reach	MEDIUM	Higgs as inflaton — specific prediction now constrained by CMB

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Radion/KK scalar	Not covered	Randall-Sundrum radion: 0.5-2 TeV, Higgs-like. No LHC signal. V2.0: RADION_SCALAR type added	LOW	Extra-dimension scalar particle extends V1.0 scalar sector

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Dark
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Scalar Boson Deep Dive Request V1.0 archived | V2.0 is the active specification as of June 2026